



LECTURE 03 : LOOPS & PATTERN PRINTING



Loops



What are Loops?

Repeat code multiple times without writing it again.



3 Types

1. `for` loop
2. `while` loop
3. `do-while` loop



1 `for` Loop

Best when you know HOW MANY times to repeat.



Syntax

```
for(initialization; condition; update) {  
    // code to repeat  
}
```



Example : Print 1 to 5

```
for(int i = 1; i <= 5; i++) {  
    cout << i << endl;  
}
```



Output

```
1  
2  
3  
4  
5
```



How `for` Loop Works

```
for(int i = 1; i <= 5; i++) {  
    cout << i << endl;  
}
```

Step by step

Step 1: $i = 1$ (initialization)
Step 2: Check $i \leq 5$? YES → Print 1
Step 3: $i++ \rightarrow i$ becomes 2
Step 4: Check $i \leq 5$? YES → Print 2
Step 5: $i++ \rightarrow i$ becomes 3
Step 6: Check $i \leq 5$? YES → Print 3
Step 7: $i++ \rightarrow i$ becomes 4
Step 8: Check $i \leq 5$? YES → Print 4
Step 9: $i++ \rightarrow i$ becomes 5
Step 10: Check $i \leq 5$? YES → Print 5
Step 11: $i++ \rightarrow i$ becomes 6
Step 12: Check $i \leq 5$? NO → Stop

Example 1 : Print Even Numbers

```
for(int i = 2; i <= 10; i += 2) {  
    cout << i << " ";  
}
```

Output

2 4 6 8 10

Example 2 : Sum of First 10 Numbers

```
int sum = 0;  
  
for(int i = 1; i <= 10; i++) {  
    sum = sum + i;  
}  
  
cout << "Sum: " << sum << endl;
```

✓ Output

Sum: 55

✓ Example 3 : Multiplication Table

```
int num = 5;

cout << "Multiplication table of " << num << endl;

for(int i = 1; i <= 10; i++) {
    cout << num << " x " << i << " = " << (num * i) << endl;
}
```

✓ Output

```
Multiplication table of 5
5 x 1 = 5
5 x 2 = 10
5 x 3 = 15
...
5 x 10 = 50
```

✓ Example 4 : Reverse Counting

```
for(int i = 10; i >= 1; i--) {
    cout << i << " ";
}
```

✓ Output

10 9 8 7 6 5 4 3 2 1

🔄 2 while Loop

Best when you DON'T know how many times to repeat.

📌 Syntax

```
while(condition) {  
    // code to repeat  
}
```

✓ Example : Print 1 to 5

```
int i = 1;  
  
while(i <= 5) {  
    cout << i << endl;  
    i++;  
}
```

✓ Output

```
1  
2  
3  
4  
5
```

✓ Example 1 : User Input Until Correct

```
int password;  
int correctPassword = 1234;  
  
cout << "Enter password: ";  
cin >> password;  
  
while(password != correctPassword) {  
    cout << "Wrong! Try again: ";  
    cin >> password;  
}  
  
cout << "Access granted!" << endl;
```

✓ Example 2 : Sum Until User Enters 0

```

int num;
int sum = 0;

cout << "Enter numbers (0 to stop):" << endl;

while(true) {
    cin >> num;

    if(num == 0) {
        break; // Exit Loop
    }

    sum += num;
}

cout << "Total sum: " << sum << endl;

```

3 do-while Loop

Runs AT LEAST ONCE, then checks condition.

Syntax

```

do {
    // code to repeat
} while(condition);

```

Example

```

int i = 1;

do {
    cout << i << endl;
    i++;
} while(i <= 5);

```

Output

```

1
2
3

```



Difference : while vs do-while

◆ while loop

```
int i = 10;

while(i <= 5) { // Condition false, never runs
    cout << i << endl;
}
```

✓ Output

(nothing)

◆ do-while loop

```
int i = 10;

do {
    cout << i << endl; // Runs once
} while(i <= 5);
```

✓ Output

10

🎮 2 Loop Control Statements

🚫 1 break - Exit Loop Immediately

```
for(int i = 1; i <= 10; i++) {

    if(i == 5) {
        break; // Stop when i is 5
    }
}
```

```
cout << i << " ";  
}
```

✓ Output

1 2 3 4

2 continue - Skip Current Iteration

```
for(int i = 1; i <= 5; i++) {  
  
    if(i == 3) {  
        continue; // Skip 3  
    }  
  
    cout << i << " ";  
}
```

✓ Output

1 2 4 5

✓ Example : Skip Even Numbers

```
for(int i = 1; i <= 10; i++) {  
  
    if(i % 2 == 0) {  
        continue; // Skip even numbers  
    }  
  
    cout << i << " ";  
}
```

✓ Output

1 3 5 7 9

Pattern Printing & Type Conversion

Part 1 : Pattern Printing (Complete Guide)

Pattern 1 : Square Pattern

```
// Print 5x5 square of stars
```

```
*****  
*****  
*****  
*****  
*****
```

Code

```
for(int i = 1; i <= 5; i++) {  
  
    for(int j = 1; j <= 5; j++) {  
        cout << "*";  
    }  
  
    cout << endl;  
}
```

Pattern 2 : Right Triangle (Increasing)

```
*  
**  
***  
****  
*****
```

Code

```
for(int i = 1; i <= 5; i++) {  
  
    for(int j = 1; j <= i; j++) {  
        cout << "*";  
    }  
  
    cout << endl;  
}
```



```
cout << endl;  
}
```



How it works

```
i=1: Print 1 star  
i=2: Print 2 stars  
i=3: Print 3 stars  
i=4: Print 4 stars  
i=5: Print 5 stars
```

★ Pattern 3 : Right Triangle (Decreasing)

```
*****  
****  
***  
**  
*
```

Code

```
for(int i = 5; i >= 1; i--) {  
  
    for(int j = 1; j <= i; j++) {  
        cout << "*";  
    }  
  
    cout << endl;  
}
```

★ Pattern 4 : Inverted Right Triangle

```
*  
**  
***  
****  
*****
```



Code

```
for(int i = 1; i <= 5; i++) {  
  
    // Print spaces  
    for(int j = 1; j <= 5-i; j++) {  
        cout << " ";  
    }  
  
    // Print stars  
    for(int j = 1; j <= i; j++) {  
        cout << "*";  
    }  
  
    cout << endl;  
}
```



Logic

Row 1: 4 spaces, 1 star
Row 2: 3 spaces, 2 stars
Row 3: 2 spaces, 3 stars
Row 4: 1 space, 4 stars
Row 5: 0 spaces, 5 stars

★ Pattern 5 : Number Square Pattern

```
1  2  3  4  5  
6  7  8  9  10  
11 12 13 14 15  
16 17 18 19 20  
21 22 23 24 25
```



Code

```
int num = 1;  
  
for(int i = 1; i <= 5; i++) {  
  
    for(int j = 1; j <= 5; j++) {  
  
        cout << num << " ";  
        num++;  
    }  
}
```

```
cout << endl;
```

```
,
```